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(hadoop@kali)-[/home/kali]
$ sudo vi word_count.txt
[sudo] password for hadoop:

(hadoop@kali)-[/home/kali]
$ sudo vi mapper.py
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(hadoop@kali)-[/home/kali]
$ sudo vi reducer.py
```

```
(hadoop@kali)-[/home/kali]
$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [kali]
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
Starting resourcemanager
Starting nodemanagers
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```
(hadoop@kali)-[~]
$ sudo mkdir exp1
```

```
(hadoop@kali)-[~]
$ sudo cp word_count.txt ./exp1
```

```
(hadoop@kali)-[~]
$ sudo chmod 777 mapper.py reducer.py
```

```
(hadoop@kali)-[/home/kali]
$ sudo unzip jar_files.zip
Archive:  jar_files.zip
  inflating: hadoop-streaming-3.3.6.jar
```

```
(hadoop@kali)-[~/exp1]
$ ls -l
total 12
-rwxrwxrwx 1 root root 157 Aug 25 12:48 mapper.py
-rwxrwxrwx 1 hadoop hadoop 459 Aug 25 12:54 reducer.py
-rw-r--r-- 1 root root 103 Aug 24 01:41 word_count.txt
```

```
(hadoop@kali)-[~/exp1]
$ hdfs dfs -mkdir /exp1
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true

(hadoop@kali)-[~/exp1]
$ hdfs dfs -put word_count.txt /exp1
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
```

```
(hadoop@kali)-[~/exp1]
$ hadoop jar $HADOOP_STREAMING -input /exp1/word_count.txt -output /exp1/output3 -mapper ~/exp1/mapper.py -reducer ~/ex
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
packageJobJar: [/tmp/hadoop-unjar11644362190866302151/] [] /tmp/streamjob3563510899481168501.jar tmpDir=null
2024-08-25 12:56:54,465 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-08-25 12:56:54,944 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-08-25 12:56:55,474 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/h
2024-08-25 12:56:56,114 INFO mapred.FileInputFormat: Total input files to process : 1
2024-08-25 12:56:56,325 INFO mapreduce.JobSubmitter: number of splits:2
2024-08-25 12:56:56,922 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1724604239961_0004
2024-08-25 12:56:56,922 INFO mapreduce.JobSubmitter: Executing with tokens: []
2024-08-25 12:56:57,437 INFO conf.Configuration: resource-types.xml not found
2024-08-25 12:56:57,437 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2024-08-25 12:56:57,830 INFO impl.YarnClientImpl: Submitted application application_1724604239961_0004
2024-08-25 12:56:58,224 INFO mapreduce.Job: The url to track the job: http://kali:8088/proxy/application_1724604239961_00
2024-08-25 12:56:58,230 INFO mapreduce.Job: Running job: job_1724604239961_0004
2024-08-25 12:57:12,822 INFO mapreduce.Job: Job job_1724604239961_0004 running in uber mode : false
2024-08-25 12:57:12,827 INFO mapreduce.Job: map 0% reduce 0%
2024-08-25 12:57:35,861 INFO mapreduce.Job: map 100% reduce 0%
2024-08-25 12:57:50,096 INFO mapreduce.Job: map 100% reduce 100%
2024-08-25 12:57:52,169 INFO mapreduce.Job: Job job_1724604239961_0004 completed successfully
2024-08-25 12:57:52,533 INFO mapreduce.Job: Counters: 54
```

File System Counters

```
FILE: Number of bytes read=180
FILE: Number of bytes written=835607
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=341
HDFS: Number of bytes written=133
HDFS: Number of read operations=11
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0
```

Job Counters

```
Launched map tasks=2
Launched reduce tasks=1
Data-local map tasks=2
Total time spent by all maps in occupied slots (ms)=40604
Total time spent by all reduces in occupied slots (ms)=11275
Total time spent by all map tasks (ms)=40604
Total time spent by all reduce tasks (ms)=11275
Total vcore-milliseconds taken by all map tasks=40604
Total vcore-milliseconds taken by all reduce tasks=11275
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Total megabyte-milliseconds taken by all map tasks=41578496
Total megabyte-milliseconds taken by all reduce tasks=11545600
Map-Reduce Framework
  Map input records=5
  Map output records=18
  Map output bytes=138
  Map output materialized bytes=186
  Input split bytes=186
  Combine input records=0
  Combine output records=0
  Reduce input groups=17
  Reduce shuffle bytes=186
  Reduce input records=18
  Reduce output records=17
  Spilled Records=36
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=412
  CPU time spent (ms)=5630
  Physical memory (bytes) snapshot=657350656
  Virtual memory (bytes) snapshot=8107618304
  Total committed heap usage (bytes)=373293056
  Peak Map Physical memory (bytes)=249540608

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Peak Map Virtual memory (bytes)=2698510336
Peak Reduce Physical memory (bytes)=172601344
Peak Reduce Virtual memory (bytes)=2712342528
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=155
File Output Format Counters
  Bytes Written=133
2024-08-25 12:57:52,544 INFO streaming.StreamJob: Output directory: /exp1/output3

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```
(hadoop@kali)-[~/exp1]
$ hdfs dfs -cat /exp1/output3/part-00000
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
Everyday      1
Everything     1
Live          1
Today         1
a             1
an            1
beginning     1
change        1
day           1
in            1
is            2
moment        1
new           1
one           1
opportunity    1
the           1
will          1
```